

REPORT: Interpretation of R-Analysis Results

Objective

The analysis aims to explore the relationship between Gross Domestic Product (GDP) and the unemployment rate (UNRATE) using a linear regression model.

Key Findings

Relationship: There is a statistically significant negative relationship between GDP and the unemployment rate. As GDP increases, the unemployment rate tends to decrease.

Model Equation: The unemployment rate (UNRATE) can be estimated using the equation:

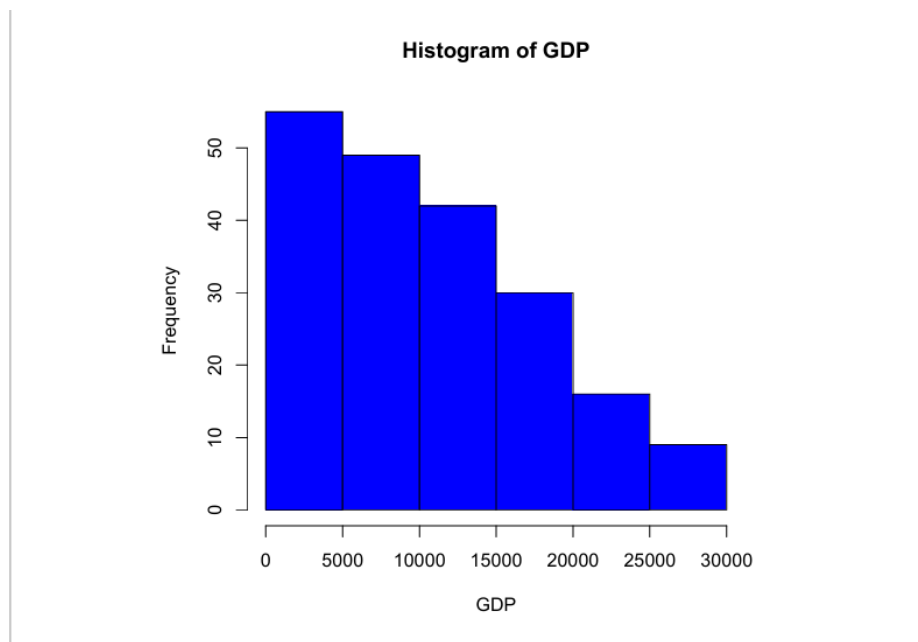
$$\text{UNRATE} = 7.162 - 0.00008947 \times \text{GDP}$$

Statistical Significance: The GDP coefficient is significant with a p-value of 2.87×10^{-7} , indicating strong evidence that GDP influences the unemployment rate.

Model Fit: The R-squared value is 0.124, meaning that GDP explains about 12.4% of the variability in the unemployment rate.

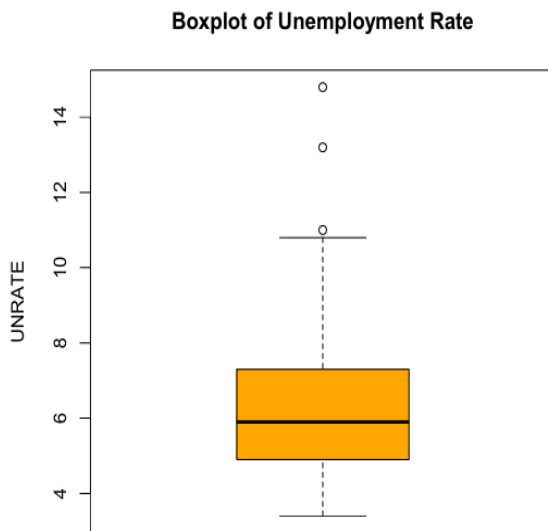
Visualisations

- **Histogram of GDP:** The distribution reflects economic disparities, where a significant portion of the data is concentrated in lower GDP ranges, potentially highlighting economic challenges in those areas.

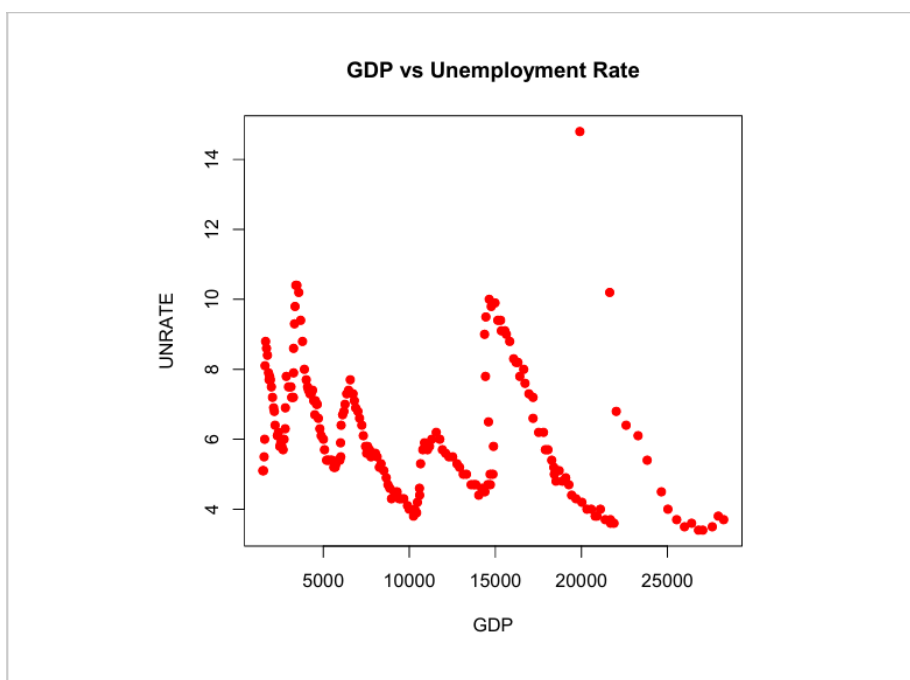


- **Boxplot of Unemployment Rate:** The boxplot effectively summarises the central tendency and variability of the unemployment rate, it shows a median

unemployment rate of around 8%, with notable variability and outliers, suggesting some regions face significantly higher unemployment challenges.



- **ScatterPlot of GDP VS Unemployment Rate:** The scatter plot illustrates a positive correlation between GDP and unemployment rates, indicating that as GDP increases, unemployment tends to decrease, suggesting a potential relationship between economic growth and job availability.



Conclusion

The analysis confirms a meaningful but modest impact of GDP on the unemployment rate, suggesting that while economic growth reduces unemployment, other factors also significantly influence unemployment trends. We can look into these other factors at a later time.